

Sachin Vasle

✉ sachinvasle99@gmail.com | ☎ +91 91116 90842 | 📍 Pune, India
🌐 linkedin.com/in/sachin-vasle | 🐙 github.com/sachinvasle99 | 🌐 sachinvasle.in

Professional Summary

DevOps/Platform Engineer with around 5 years of experience running **hybrid Kubernetes** across managed cloud and bare-metal on-premises infrastructure, together with the **AI and GPU infrastructure** layered on top. Has delivered cloud migration in **both directions** — on-premises to AWS EKS at one organization, and EKS back onto self-operated Kubernetes at the next, including the private cloud build-out beneath it. Operates self-hosted LLM inference on on-premises GPU compute, and is the sole designer and developer of **Krista XRay**, an AI-powered SRE root cause analysis platform built on a deterministic, read-only core.

Work Experience

Krista AI (*Enterprise AI Automation Platform*) | DevOps Engineer

Oct 2024 – Present

- **Cloud repatriation and private cloud build-out.** Led the migration of production workloads **off AWS EKS onto self-operated Kubernetes**, standing up the data-centre platform that replaced the managed control plane — compute, storage classes, load balancing, container registry, secrets and ingress — enabling in-place upgrades on owned hardware. Driven by cost and data-residency requirements.
- **Kubernetes lifecycle management.** Control-plane and worker provisioning, **etcd** health and backup, rolling upgrades and certificate rotation, with deep troubleshooting across the **API server, scheduler, kubelet, CoreDNS and CNI** on both **RKE2 bare metal** and **Amazon EKS**. Operate the controller layer — **Sealed Secrets, KEDA** and the **Prometheus Operator** — covering CRD lifecycle, operator upgrades and reconciliation failures.
- **AI and GPU infrastructure.** Operate **self-hosted LLM inference on on-premises GPU compute**, covering model serving, GPU utilization, inference performance, context management and tool-calling, and built **Krista XRay** on top of it. Extending into Kubernetes orchestration of GPU workloads.
- **CI/CD and GitOps delivery.** **Jenkins via JCasC**, with controllers, plugins, credentials and jobs defined as YAML in Git, alongside GitLab CI and Bitbucket Pipelines running in-cluster, **JFrog Artifactory** for artifact promotion, Helm and Kustomize for packaging, and **GitOps on ArgoCD** with automated rollback.
- **Hybrid networking.** CNI and CoreDNS on bare metal, ingress controllers and load balancing, **network policies** and multi-tenant namespace isolation, plus the firewall and routing configuration required for hybrid connectivity.
- **Observability and on-call.** **Prometheus, Grafana, SigNoz, OpenTelemetry** and **ELK/EFK** for metrics, logs and traces, with dashboards and alerting used for incident response and escalation through **PagerDuty**.
- **Pipeline security.** **Trivy** image and IaC scanning in CI to gate known CVEs before deployment; **Sealed Secrets** and the **External Secrets Operator** keeping secrets encrypted at rest in Git, with **HashiCorp Vault** and cloud Secrets Manager as the source of truth. GitOps enforces reviewed commits as the sole change path, with audit trail and rollback, under RBAC and least-privilege IAM, and production read-only by default.
- **Compliance and audited environments.** **SOC 2** change management with a reviewable trail, **HIPAA** encryption in transit and at rest with tenant separation, and **data residency** delivered through dedicated single-tenant environments in the customer region combined with self-hosted inference.
- **AWS infrastructure as code.** Multi-region workloads with VPCs, subnets, IAM and node groups in reusable **Terraform** and **Terragrunt** modules, **Ansible** for node build-out, and cost optimization through rightsizing, storage lifecycle policies and idle-resource cleanup.
- **Database performance.** Migrated ephemeral-environment PostgreSQL volumes onto **RAM-backed storage**, removing disk I/O as the bottleneck and substantially reducing environment provisioning and test-cycle time.

Revent Labs | Software Engineer (SDE-1) — Cloud & Platform

Dec 2022 – Sep 2024

- **On-premises to EKS migration.** Planned and executed the move of workloads from on-premises infrastructure onto **Amazon EKS**, provisioning networking, IAM, node groups and the delivery path in the target account ahead of cutover.
- **Region-portable infrastructure as code.** Reusable **Terraform** modules composed with **Terragrunt**, allowing an environment to be replicated into another region from the same source and enabling **rapid multi-region failover**.
- **AWS breadth.** VPC, IAM, EC2, ELB, ASG, EKS, ECS, ECR, EBS, S3, RDS and Aurora, Route 53, Lambda, CloudWatch, SQS, SNS and Secrets Manager, architected for high availability, security and scale.
- **GCP breadth.** VPC, Compute Engine, IAM, firewalls, **GKE**, Artifact Registry, GCS, Cloud SQL, Cloud Logging, Pub/Sub, Cloud Functions, load balancing and Secret Manager.
- **Database migrations.** Led **Amazon RDS to Aurora Serverless**, implemented a **global RDS cluster**, and migrated **MongoDB to Atlas** for scalability and performance headroom.
- **Kubernetes delivery and autoscaling.** EKS managed through **FluxCD** and **Helmfile**, increasing deployment frequency, with **HPA, VPA** and **KEDA** tuning cluster behaviour to actual load.
- **CI/CD and quality gates.** **GitHub Actions** pipelines integrated with **Jira** for traceability and **SonarQube** for code quality gates.
- **Monitoring and secrets management.** CloudWatch, Grafana, **Loki** and **Promtail** and the EFK and PLG stacks, with alerting configured to reduce response time; **Vault, Sealed Secrets** and the **External Secrets Operator** integrated with AWS and GCP Secrets Manager to keep credentials out of manifests.

- **Automation and legacy migration.** Ansible playbooks, roles and collections driven from GitHub Actions for end-to-end deployment and testing, plus migration of **RightScale RightScripts** to equivalent Terraform configurations and Ansible roles.
- **Cost optimization.** Rationalized Route 53, **Transit Gateway** and **AWS Organizations** usage to reduce spend.

DXC Technology | Associate Software Engineer

Jul 2021 – Nov 2022

- **EKS cluster management at scale.** Operated production clusters carrying **100+ workloads** across enterprise client environments, managing resource utilization and platform stability, and served as escalation point for critical Kubernetes production issues.
- **Helm delivery and image builds.** Developed and maintained **Helm** charts for application deployment onto EKS, and designed **multistage Dockerfiles** that minimize runtime layers and ensure image parity across development, test and production.
- **Ingress and network policy.** Configured **Nginx** and **Traefik** ingress controllers for application traffic, with **Cilium** and **Calico** network policies enforcing workload isolation.
- **Observability and alerting.** Built and maintained **Grafana** alerts covering system health, surfacing infrastructure and application issues to the SysOps team ahead of user impact.
- **Cost optimization and automation.** Reduced AWS spend across multiple client projects using **Kubecost** for per-workload visibility on EKS, alongside traffic optimization via Traefik; developed reusable **Bash** tooling to standardize testing procedures.

Projects

- **Krista XRay — AI-Powered SRE Root Cause Analysis Platform** (sole designer and developer, in production). Performs automated root cause analysis for production incidents by querying **nine telemetry sources in parallel** — Kubernetes, Prometheus, traces, logs, PostgreSQL, Redis and cloud APIs — normalizing them onto a single timeline, detecting changes within the incident window, ranking probable causes and reporting the leading candidates with evidence cited per claim. Python backend across **63 modules** with a React/Vite frontend; asynchronous **FastAPI** service with Pydantic-typed models, deployed to Kubernetes under RBAC-scoped service accounts.
- **Deterministic analysis core.** All numeric analysis is computed in Python and the model only interprets the results, preventing fabricated metrics. All **16 agent tools are read-only** and service accounts RBAC-scoped, leaving no code path from model output to a cluster write; collectors fail explicitly rather than silently dropping a source.
- **Hypothesis ranking.** A dependency graph scores candidate causes before the model is invoked, so the agentic loop evaluates a ranked shortlist rather than raw telemetry, with change detection over the same window resolving a significant share of incidents directly.
- **Multi-provider inference.** Abstraction over the hosted **Claude API** and **self-hosted models on on-premises GPUs**, with **tiered model routing** and **prompt caching** of static system and tool definitions to control cost and latency; an **MCP bridge** exposes SigNoz, Grafana and PMM as typed tools.

Technical Skills

Kubernetes:	RKE2 on bare metal, Amazon EKS, GKE, etcd, CNI, CoreDNS, kube-apiserver / scheduler / kubelet, Helm, Kustomize, ingress (Nginx, Traefik), network policy (Cilium, Calico), RBAC, multi-tenant isolation, HPA / VPA / KEDA, Operators & CRDs
Cloud:	AWS (VPC, IAM, EC2, ELB, ASG, EKS, ECS, ECR, EBS, S3, RDS & Aurora, Route 53, Lambda, CloudWatch, SQS, SNS, Secrets Manager, Transit Gateway), GCP (GKE, Compute Engine, VPC, IAM, Artifact Registry, GCS, Cloud SQL, Pub/Sub)
IaC & Config:	Terraform, Terragrunt, Ansible (playbooks, roles, collections), Helm, Kustomize, YAML/JSON
CI/CD & GitOps:	Jenkins (JCasc), GitHub Actions, GitLab CI, Bitbucket Pipelines, ArgoCD, FluxCD, Helmfile, JFrog Artifactory, SonarQube, Jira, Git
Observability:	Prometheus, Grafana, SigNoz, OpenTelemetry, Loki & Promtail, ELK/EFK, PLG, CloudWatch, distributed tracing, dashboards, alerting, PagerDuty on-call
Security & Compliance:	Trivy (image & IaC scanning), HashiCorp Vault, Sealed Secrets, External Secrets Operator, RBAC, least-privilege IAM, network policy, SOC 2, HIPAA, data residency, audit trail
AI / GPU:	Self-hosted LLM serving on on-premises GPUs, inference performance & GPU utilization, agentic tool-calling loops, MCP server development, context management, prompt caching, tiered model routing, read-only guardrails
Programming:	Python (FastAPI, Pydantic, httpx, asyncio, boto3, Kubernetes client, Anthropic SDK), Bash/Shell, Groovy, JavaScript/React, REST API design
Data:	PostgreSQL (incl. RAM-backed storage), MySQL, MongoDB & Atlas, Aurora Serverless, Redis, ClickHouse, Elasticsearch
OS & Infrastructure:	Linux (RHEL, Ubuntu), Docker (multistage builds), containerd, bare-metal & data centre infrastructure, DNS, DHCP, Nginx, load balancing, Kubecost

Education & Certifications

B.E. (Computer Science), Dr. D. Y. Patil Institute of Technology, Pimpri, Pune — CGPA 7.52/10

2017 – 2021

AWS Certified Solutions Architect – Associate (SAA-C03)

Jun 2024 – Jun 2027

Certified Kubernetes Administrator (CKA), Linux Foundation

Feb 2022 – Feb 2025